Greecon
SH.P.K

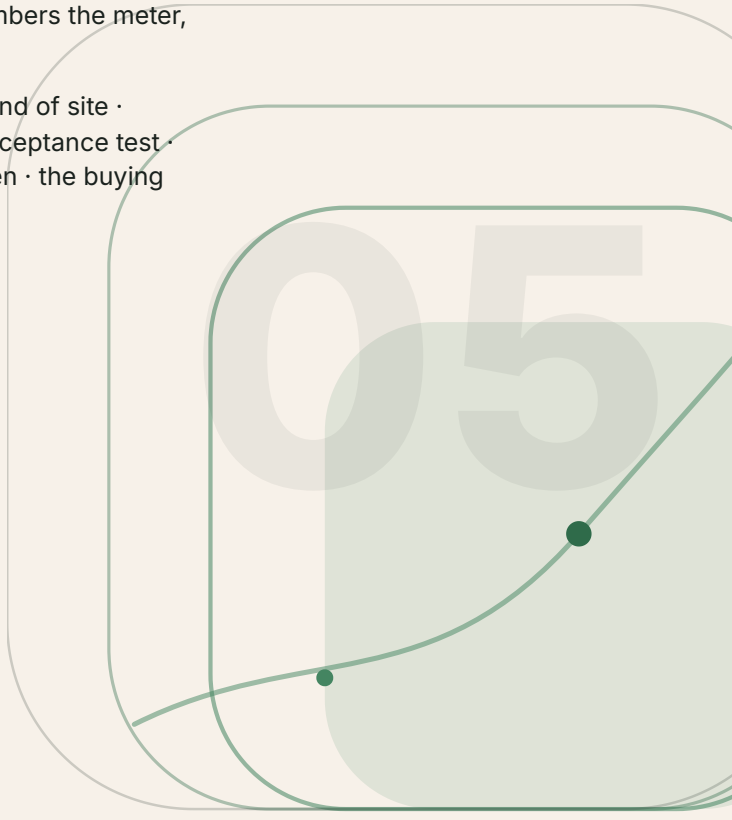
05

HARDWARE SHEET HW05 · APPROVED MODEL

Three-phase meter: Eastron SDM630

The meter at the grid connection, on the solar output and, where wanted, on the site's own consumption.

QUALIFIES	Approved model. The Eastron SDM630 with Modbus. Greecon reads it by its own register table and recognises it on the network, so another three-phase meter does not take its place.
USED IN	Kits H and S (H hotel or guesthouse, S solar plant, F farm or greenhouse, W water tank and pump station)
FOR	The project manager who buys and numbers the meter, and the electrician who fits it
COVERS	How it qualifies · what it does at each kind of site · settings · fitting · in the platform · the acceptance test · what goes wrong · what is not yet proven · the buying record



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HW05

Three-phase meter: Eastron SDM630

The Eastron SDM630 is the three-phase meter of every kit that measures electricity. Fitted where the supply enters the site, it gives both what is bought from the grid and what is sent to it. Fitted on the inverter's output, it gives what the solar plant produces. Greecon knows its registers, so nobody types one, and the box recognises it on the network.

At a glance

WHAT IT DOES	Measures electrical power on three phases. One at the grid connection (unit 1) gives both Grid import and Grid export; one on the inverter's AC output (unit 2) gives Solar production; a third, if wanted, on the site's own consumption (unit 4). Fitted by the electrician.
HOW IT QUALIFIES	**Approved model.** The Eastron SDM630 with Modbus. Greecon reads it by its own register table and recognises it on the network, so another three-phase meter does not take its place.
MAKE AND MODEL	Eastron SDM630 Modbus. The MCT version comes with three clamp-on current sensors for up to 100 A a phase; for more than 100 A, the ordinary SDM630 with 5 A current transformers
NUMBER ON THE CABLE	1 as the grid meter, 2 as the solar meter, 4 as the load meter. Set on its own display, at the office.
FITTED BY	The electrician, inside the distribution board, under the customer's lock-out. The project manager sets its number at the office and runs the data cable to the board.
PLANNING ESTIMATE (EUR)	90 to 180 each. Not a quote.
USED IN	Kits H and S. The table under *At each kind of site* says what it does at each.

How it qualifies

It is an approved model: the make and the model are named, and no other three-phase meter takes its place.

It qualifies if:

- ☐ It is an Eastron **SDM630** with Modbus (RS-485). The model on its label is the model on the quote.
- ☐ Its version suits the current. Up to 100 A a phase: the **MCT** version, which comes with three clamp-on current sensors. Above 100 A: the ordinary SDM630 with 5 A current transformers, bought to suit the supply.

☐ The maker's manual and Modbus protocol document came with it. Versions differ: read the one in its box.

STOP

Do not buy another make of three-phase meter because it "also speaks Modbus". The box would not recognise it and the template's registers would not fit it. It could only be entered by hand, from its own register table, as equipment Greecon does not know.

At each kind of site

Kind of site	In its kit	How many	What it does there
H Hotel or guesthouse	IN THE KIT	2	Two. The grid meter (number 1), where the supply enters, gives Grid import and Grid export. The solar meter (number 2), on the inverter's AC output, gives Solar production.
S Solar plant with grid connection	IN THE KIT	2, or 3	Two, as at a hotel: the grid meter (1) and the solar meter (2). A third (number 4) on the feed to the loads if the owner wants the site's own consumption measured and not worked out.
F Farm or greenhouse with irrigation	NOT USED	None	A farm kit has no energy meter. One can be added to any site from Add equipment Greecon already knows, and then this sheet applies.
W Water tank and pump station	NOT USED	None	The kit measures the pump's electricity with the single-phase SDM230 (sheet HW06). A three-phase pump needs this meter in its place: remove the template's pump meter and add Eastron SDM630 on the site's own consumption (number 4) from Add equipment Greecon already knows.

What Greecon reads from it

One number: the **total active power**. It is input register 52 (counted from zero; the manual's 30053), a 32-bit floating-point number, high half first, in watts. Greecon shows it in kW. The template fills the register in: nobody types it.

Fitted as	Reading in Greecon	How it is made from the power
Grid meter (number 1)	**Grid import**, kW	The power while it is positive: energy coming in
	Grid export, kW	The power while it is negative, shown as a positive number: energy sent out
Solar meter (number 2)	**Solar production**, kW	The size of the power, whichever way the clamps point
Load meter (number 4)	**Energy consumption**, kW	The size of the power

That is why one grid meter gives both import and export, and why a clamp fitted the wrong way round swaps them. Greecon notices a reversed grid meter on its own at night or in cloud, because energy cannot leave a site that nothing is producing for: the energy picture says so at once, and after five minutes a **Check today** alert opens, *The grid meter may be fitted the wrong way round*. At midday it cannot tell, which is why the direction check stays.

Settings, and where they are made

They differ between versions, so they are read from the meter's own display and never assumed. Greecon sets them at the office; on site you only check them, and photograph the display.

Setting	Value	Where
Unit number	1 on the grid meter, 2 on the solar meter, 4 on the load meter. A meter is delivered as 1, so each is numbered alone, before it joins the cable	On its display, at the office
Speed	9600 baud, which is how it is delivered: check it	On its display, at the office
Parity and stop bits	No parity, 1 stop bit	On its display, at the office
CT ratio	Only on a meter with 5 A current transformers: the ratio of the transformers fitted (for example 250 for a 250/5 A transformer), so that the meter shows the real power	In the meter's menu, on site, by the electrician

A meter's display works only when its voltage terminals have power, and powering a meter on a bench is mains work. The electrician makes up the office's bench lead. The project manager never wires mains.

Fitting

ELECTRICIAN Inside a distribution board, with the customer's lock-out on. The project manager watches, photographs and writes it down.

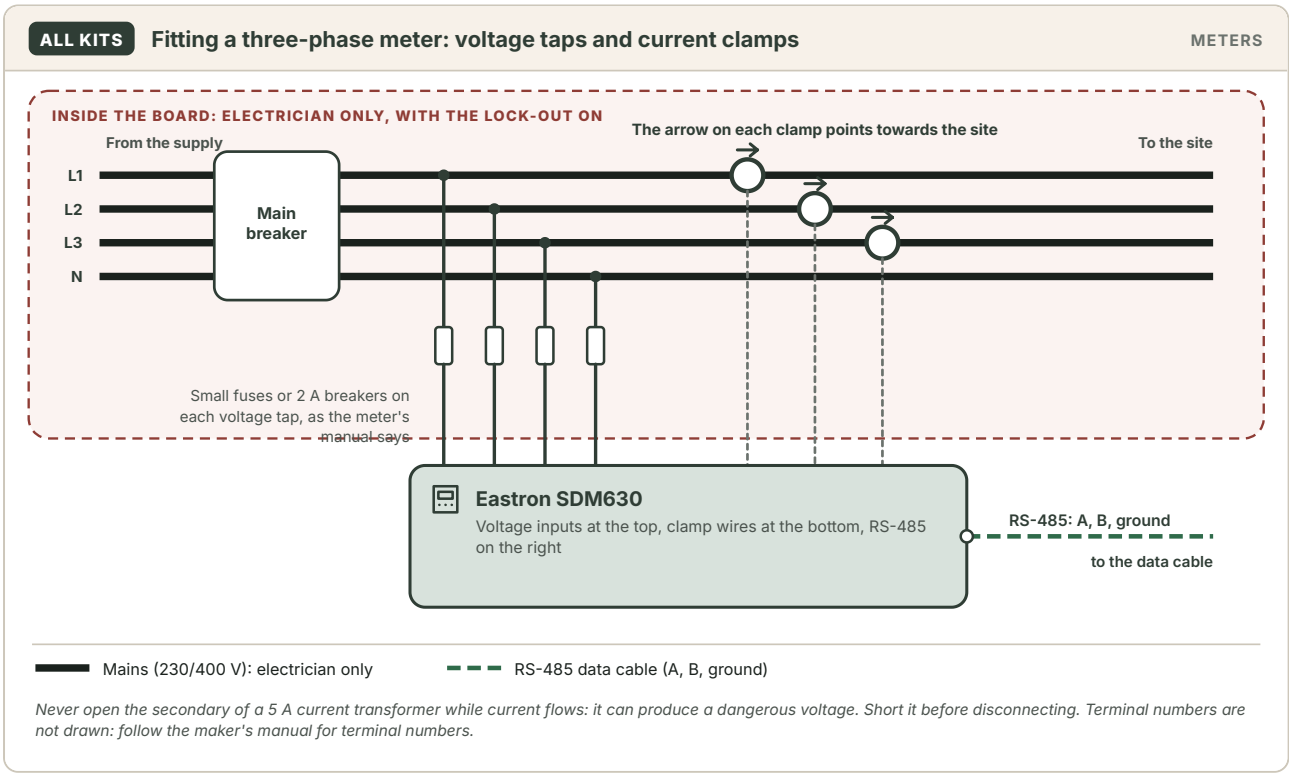


Figure 4-2. Fitting a three-phase meter: fused voltage taps and three current clamps, arrows towards the site. The electrician's work.

- [ ] **Clamps point towards the site.** On the grid meter this makes import positive. On the solar meter the clamps point away from the inverter, towards the board. Solar is read as the size of the power, so a reversed solar clamp does not read zero, but fit it the right way anyway.
- [ ] **Voltage taps** have small fuses or 2 A breakers on each tap, as the meter's manual says.
- [ ] **Never open the secondary of a 5 A current transformer while current flows:** it can produce a dangerous voltage. Short the secondary before disconnecting.
- [ ] The electrician connects the data cable at the meter, because its terminals are inside the board.
- [ ] **Photograph** the wiring, the label (model and serial number) and the display with its settings.

In the platform

- **Finding it.** In step 4 of the setup list, **Where each device is on the network**, press **Find equipment**. The box recognises an Eastron meter by what it says: a first-phase voltage that is a mains voltage, and a power. Found at the number the kit gives it, its address is filled in and the line says **Set**, for example *unit 1: an Eastron meter, reading 1.80 kW at 231 V*.
- **Found, but waiting for you.** A meter at another number than the kit's, or at another address than the one it has, says **Found** and is set only when you press **Use this address**.
- **A single-phase meter where this one was planned.** The line says *It is a single-phase SDM230, not the three-phase SDM630 planned.* If that is what the site has, press **Use this address** and Greecon reads it as an SDM230 (sheet HW06). If a three-phase meter was meant, do not press it: the electrician changes the meter.
- **The direction check** (step 9, **Grid meter direction checked**, at sites with a grid meter): with the sun down or the solar switched off, switch on a known load. Grid import must go up. Then press **I checked this**.

Acceptance test

At the office:

- [ ] The number and the speed are read from the meter's display, and photographed.
- [ ] On the desk, behind the converter, the box finds it at its number and reads its power:

```
greecon-edge find --host 192.168.1.50
greecon-edge modbus-test --host 192.168.1.50 --unit 1 --type input --address 52 --data-type
float32
```

The second prints the raw registers and the value, in watts.

On the day, as part of commissioning:

Check and how	Pass when	Greecon shows	Real	Result
The meter's display against Greecon. Read the display (kW or W) and Greecon at the same time.	Within 5 percent, or 1 kW on a small load			☐ Pass ☐ Fail
Grid direction. A known load on (a kettle or a heater), the sun down or the solar off.	Import rises; export does not			☐ Pass ☐ Fail
Solar at night.	Zero, or a few watts			☐ Pass ☐ Fail
Solar at midday. Against the inverter's own display.	Within about 5 percent			☐ Pass ☐ Fail

What goes wrong

What you see	Likely cause	What to do
Import and export are swapped (export rises when a load is switched on)	The grid meter's clamps are the wrong way round	The electrician turns them, with the lock-out on. Repeat the direction check
The solar meter reads zero at midday	A clamp closed on the wrong cable, a breaker open upstream, or an inverter limited to zero export	Compare with the meter's own display. Ask the electrician to confirm that the clamps are on the inverter's output phases. On a zero-export plant, midday solar can honestly be low: check the inverter's display
Nothing that looks like an Eastron meter answers at unit 2	No power at the meter, the number or the speed on its display is not the kit's, or its A and B wires	Read its display against the install card. Check its two data wires. A meter powered from the grid is dark in a power cut: that is expected
Something answers … but not clearly enough to tell whether it is the three-phase SDM630 planned	One of its answers was lost on the cable	Press **Look again**. If it is the same, check the speed and the A and B wires
The reading is far from the display on a meter with current transformers	The CT ratio is not set in the meter	The electrician sets the ratio in the meter's menu. Greecon reads what the meter shows

More is in the runbooks (Project Manager Field Manual, PDF 02): R2 (a device does not answer), R3 (a reading looks wrong), R15 and R16 (Find equipment).

Not yet proven

- This software has not yet read a real SDM630. The register is from the maker's Modbus protocol document, and everything is tested against a pretend meter that answers as that document says. The first real kit is rehearsed on a desk, and the first reading at a site is compared with the meter's own display.

- The prices are planning figures from shop listings seen in October 2026, not quotes.

Buying record

Ask three suppliers for a quote and keep the quotes in the shared drive. Buy one, pass the acceptance test with it, and only then buy more. For every part that is bought the hardware register keeps the make and model, the supplier, the date, the serial number or the batch, and the invoice.

	Quote 1	Quote 2	Quote 3
Supplier			
Make and model offered			
Price (EUR, without VAT), and delivery			
Meets every line of *It qualifies if*	☐ Yes	☐ Yes	☐ Yes

Bought	Answer
Make, model and version	
Supplier, date and invoice	
Serial number or batch	
Acceptance test passed on (date), by	